P-W-158c
January 2, 1964
SUPERSEDING
Fed. Spec. P-W-158b
May 20, 1959

FEDERAL SPECIFICATION

WAX, GENERAL PURPOSE, SOLVENT TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of general purpose waxes suitable for use on furniture, painted surfaces and floors, except asphalt, vinyl and rubber composition floors (see 6.1).

1.2 Classification.

1.2.1 *Types.* The waxes shall be of the following types, as specified (see 6.2):

Type I—Liquid
Type II—Paste

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-C-96—Cans, Metal, 28 Gage and Lighter.
PPP-D-729—Drums; Metal, 55 Gallon (for Shipment of Noncorrosive Material).
PPP-P-704—Pails; Shipping, Steel (1 through 12 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Federal Hazardous Substance Labeling Act.

(Copies may be obtained from the U. S. Department of Health, Education and Welfare, Food and Drug Administration, Washington, D. C. 20407.)

American Society For Testing and Materials (ASTM) Designations:

D 56-61—Method of Test for Flash Point by Tag Closed Tester.
D 217-60 T—Method of Test for Cone Penetration of Lubricating Grease (Tentative).
D 1386-59—Method of Test for Acid Number (Empirical) of Natural Waxes.
D 1387-59—Method of Test for Saponification Number (Empirical) of Natural Waxes.
D 1542-60—Method of Qualitative Tests for Rosin in Varnishes.
E 28-58 T—Method of Test for Softening Point by Ring and Ball Apparatus (Tentative).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 **Material.** Type I and type II general purpose wax shall consist essentially of natural or synthetic waxes, or a blend thereof, in a volatile organic solvent.

3.2 **Physical state.**

3.2.1 *Type I.* Type I liquid wax shall be a free flowing liquid mixture of suitable consistency to permit ease of application at 20° to 30°C.

3.2.2 *Type II.* Type II paste wax shall be a soft uniform paste at 30° ± 1°C. having a cone penetration range of 60 to 70, when tested as specified in 4.4.1.

3.3 **Nonvolatile matter.**

3.3.1 *Type I.* Type I liquid wax shall contain not less than 11.0 percent by weight of nonvolatile matter when tested as specified in 4.4.2. The characteristics of the nonvolatile matter shall conform to the requirements specified in table I.

3.3.2 *Type II.* Type II paste wax shall contain not less than 20.0 percent by weight of nonvolatile matter when tested as specified in 4.4.2. The characteristics of the nonvolatile matter shall conform to the requirements specified in table I.

TABLE I.—*Nonvolatile matter requirements*

	Types I and II	
	Minimum	Maximum
Softening point, minimum	77°C.	80°C.
Acid value, maximum	0	10
Saponification value	20	100
Ash content, percent, maximum	0.0	0.5
Rosin	None	None

3.4 **Volatile organic solvent.** The volatile organic solvent (vehicle) shall consist of turpentine or volatile petroleum distillates, or any mixture thereof. The flash point of the volatile solvent shall be not less than 28°C. when tested as specified in 4.4.8.

3.5 **Abrasive.** Type I and type II waxes shall show no evidence of abrasive matter when tested as specified in 4.4.9.

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3.6 Stability.

3.6.1 *Type I.* Type I wax shall show not more than 5.0 percent by volume of settling of suspended material or liquid separation when tested as specified in 4.4.10.1.

3.6.2 *Type II.* Type II wax shall show no separation of liquid from the paste form when tested as specified in 4.4.10.2.

3.7 Film properties (type I and type II). The wax shall dry to a film that polishes easily to a hard lustrous surface. The wax shall not be so highly colored as to stain the surface, and shall dry to touch within 45 minutes to a semi-transparent non-tacky film which shall show no sign of whiteness when tested as specified in 4.4.11.

3.8 Performance test. Type I and type II wax shall meet the performance test specified in 4.4.12.

3.9 Workmanship. The wax shall be a uniform, homogeneous mixture in liquid or paste form. The quality of the finished product shall conform to the levels established herein. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQLs) specified in section 4 for any examinations and tests of materials, components and end items, as applicable.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and tests. Sampling for inspection and tests shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated. For purposes of sampling for inspection, a lot shall consist of all material of the same type, made from the same components and submitted for inspection at one time.

4.2.1 *Inspection of components and materials.* The supplier shall be responsible for insuring that the quality assurance provisions for components or materials shall be in accordance with this specification and with referenced subsidiary specifications to the extent applicable. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.1.5. Random samples shall be drawn from each lot of end items of the same type and size for inspection of visual, contents and preparation for delivery defects. The lot size for purposes of determining the sample size, in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of the same capacity, consisting of wax manufactured in one batch of the specified type, for the examination in 4.3.1.1, 4.3.1.2 and 4.3.1.3, and in units of shipping containers for examination under 4.3.1.4.

4.3.1.1 *Examination of the primary container.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Construction	Not type and class specified (see 5.1).
Closure	Not type specified; imperfect.
Exterior coating	Not as specified.
Workmanship	Dented, abraded, dirty, split, punctured; any other defect affecting serviceability; any seepage of contents.
Markings	Missing, incomplete, incorrect; not as specified.

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4.3.1.2 *Examination of contents of the primary container.* The sample unit for this examination shall be the contents of one filled primary container.

Examine	Defect
Form and workmanship	Not uniform, not homogeneous, contains lumps, liquid separates out, presence of foreign material.
	Not type specified.

4.3.1.3 *Examination of the end item for net contents.* The sample unit for this examination shall be one filled primary container. The lot shall be unacceptable if the average net content is less than specified or indicated for all sample units examined.

4.3.1.4 *Examination of the preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship and contents comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers, fully prepared for delivery, shall be examined for closure defects. This examination does not apply when the original container is the shipping container.

Examine	Defect
Markings	Incorrect, incomplete, illegible, omitted; of improper size, sequence, or method of application; not in accordance with contract requirements.
Material	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps; loose or inadequate sealing, strapping, or stapling.
	Bulged or distorted containers.
	Containers not internally and externally lined, as applicable.
Contents	Number of primary containers per shipping container is more or less than required.

4.3.1.5 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	L6	1.5
4.3.1.3	L4	—
4.3.1.4	L4	4.0

4.3.2 *Testing of the end item.* The end item shall be tested for the characteristics specified in table I in accordance with test methods specified in table II. A 5-pint composite sample of type I wax shall be made by taking 5 one-pint samples from 5 separate containers selected at random throughout the lot, mixed thoroughly, labelled, dated, and used for tests. Similarly 5 eight-ounce samples of type II wax shall be composited for tests. There shall be no failure to meet the specified requirements on the composite samples.

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TABLE II.—*Instructions for testing end item (composite)*

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit composite sample[1]	Results reported as	
	Requirement	Test method	Individ. unit	Lot aver.		Pass[2] or fail	Numerically to nearest
Consistency (type II)	3.2.2	4.4.1		X	Average of 2		1.0 unit
Nonvolatile matter (type I)	3.3.1	4.4.2			Average of 2		0.1 percent
Nonvolatile matter (type II)	3.3.2	4.4.2			Average of 2		0.1 percent
Softening point of nonvolatile matter	Table I	4.4.3			Average of 2		1.0°C.
Acid value of nonvolatile matter	Table I	4.4.4			Average of 2		1.0 unit
Saponification value of nonvolatile matter	Table I	4.4.5			Average of 2		1.0 unit
Ash content	Table I	4.4.6			Average of 2		0.1 percent
Rosin	Table I	4.4.7			1	X	
Flash point of solvent	3.4	4.4.8			Average of 2		—
Abrasives	3.5	4.4.9			1	X	1.0°C.
Stability (type I	3.6.1	4.4.10.1			Average of 2		—
Stability (type II)	3.6.2	4.4.10.2			1	X	0.1 percent
Film properties	3.7	4.4.11			1	X	
Performance test	3.8	4.4.12			1	X	

[1] Test reports shall include all values which results are based.
[2] If failure is indicated report either description of failure or numerical point of failure as applicable.

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4.4 Test procedure.

4.4.1 *Consistency (type II)*. Determine the cone penetration of the wax, in block form, in accordance with ASTM Designation D 217-60 T.

4.4.2 *Nonvolatile matter.* Accurately weigh 3 to 5 grams of the sample in a weighed flat-bottom dish about 8 cm. in diameter and provided with a tightly fitted cover. Remove the cover and place sample over a steam bath until the bulk of volatile matter has evaporated, and then place in an oven at 100° to 110°C. for two hours. Cool, replace cover and weigh. Repeat heating for another hour and reweigh. If sample is losing weight, replace in an oven overnight; if sample appears to be gaining weight, heat overnight in an atmosphere of carbon dioxide. Cool sample, replace cover and make final weighing. From weight of the dried residue and the weight of the original sample, calculate the percent nonvolatile matter.

4.4.3 *Softening point.* The dried residue shall be obtained in accordance with 4.4.2, except that the sample shall be sufficient in quantity to result in an ample residue for determining the softening point in accordance with ASTM Designation E 28-58 T.

4.4.4 *Acid value.* The acid value shall be determined in accordance with ASTM Designation D 1386-59 T.

4.4.5 *Saponification value.* The saponification value shall be determined in accordance with ASTM Designation D 1387-59 T.

4.4.6 *Ash content.* The dried residue shall be obtained in accordance with 4.4.2, except that the sample shall be sufficient in quantity to result in an ample residue for determining the ash content as follows: Weigh accurately from 2 to 5 grams of nonvolatile material in a tared porcelain crucible. Burn the combustible matter off slowly and ignite to constant weight. Calculate residue as percentage of ash.

4.4.7 *Rosin.* The wax shall be tested for the presence of rosin in accordance with the Liebermann-Storch Test in ASTM Designation D 1542-60.

4.4.8 *Flash point of the volatile organic solvent.* Place 80 to 100 grams of the sample wax in a distilling flask and distill off the solvent. The flash point of the distillate shall be determined in accordance with ASTM Designation D 56-61.

4.4.9 *Abrasive properties.* Spread a thin film of wax over a microscope slide. Place another microscope slide over the film of polish. Press the two slides together using a slight pressure and at the same time sliding one glass slide over the other. The movement between the glass slides shall be smooth without any indication of the presence of abrasive.

4.4.10 *Stability.*

4.4.10.1 *Type I.* A graduated glass cylinder, about one inch in diameter, shall be filled with a thoroughly mixed sample of the liquid wax to a height of ten inches. The cylinder shall be glass stoppered and left undisturbed at a temperature of 30° ± 1°C. for 24 hours followed by 24 hours at 25° ± 1°C. The degree of settling of the suspended matter in the vehicle of liquid separation shall be noted.

4.4.10.2 *Type II.* A sample of the paste wax shall be examined after standing in the closed container at 30° ± 1°C. for 24 hours followed by 24 hours at 25° ± 1°C. There shall be no separation of solvent from the wax.

4.4.11 *Properties of the film.* Take a clear glass plate of a thickness not more than 1/4 inch. Using a small sponge or cloth spread a thin layer of wax uniformly over 40 square inches of the surface of the plate. Allow to dry in a horizontal position for 45 minutes at a temperature of 20° to 25°C. and a relative humidity of 50 to 65 percent. Polish

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the plate with a dry cloth. The film shall be free from tackiness and shall show no signs of whiteness. When the film is viewed through the underside of the glass plate and by bright daylight transmitted through it, the film shall be from semi-transparent to clear and practically colorless.

4.4.12 *Performance tests.* The following tests shall be performed at a temperature of 20° to 30°C. Using a small amount of the wax on a soft, clean cotton cloth, apply two coats to clean pieces of linoleum and varnish-sealed wood. Polish each coat with a clean, dry, cloth using moderate pressure. The wax shall spread easily to a thin film and when dry shall have a lustrous finish free from tackiness. The polished linoleum and varnish-sealed wood pieces shall be stroked firmly with a clean cloth immediately following the polishing operation. Visible streaks produced in this manner indicates smearing and streaking. The film shall not crack when the piece of linoleum with the treated surface on the outside is bent over a bar or mandrel three inches in diameter through an arc of 180°. (The linoleum used in the test shall be Official Test Linoleum distributed by the Chemical Specialties Manufacturer's Association (see 6.3).

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Type I.*

5.1.1.1.1 *One-gallon containers.* One gallon of the wax shall be filled into a one-gallon oblong can conforming to type V, class 4, of PPP-C-96 having a screw-cap closure with an inner seal and provided with a bridge type handle. Closure of the can shall be as specified in the Appendix of PPP-C-96.

5.1.1.2 *Type II.*

5.1.1.2.1 *One-pound can and four-pound can.* One pound or four pounds of wax shall be packaged in a round can conforming to type V, class 2 of PPP-C-96. Closure shall be as specified in the Appendix of PPP-C-96.

5.1.2 *Level C.* Type I and type II wax shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Type I.*

5.2.1.1.1 *One-gallon containers.* The wax, packaged as specified in 5.1.1.1.1, shall be packed in fiberboard shipping containers in accordance with the exterior packing requirements for overseas shipment in the Appendix of PPP-C-96.

5.2.1.1.2 *Five-gallon containers.* Five gallons of the wax shall be filled into a metal pail conforming to type I, class 8 of PPP-P-704, having a screw cap closure with an inner or outer seal. Each pail shall have exterior and interior protective coatings. Color of the exterior coating shall acceptably match color X24087 of Fed. Std. No. 595.

5.2.1.1.3 *Fifty-five gallon containers.* Fifty-five gallons of the wax shall be filled into a metal drum conforming to type II of PPP-D-729.

5.2.1.2 *Type II.*

5.2.1.2.1 *One-pound can and four-pound can.* The wax, packaged as specified in 5.1.1.2.1, shall be packed in fiberboard shipping containers in accordance with the exterior packing requirements for overseas

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shipment in the Appendix of PPP-C-96, except that the four-pound can shall be packed in the same manner as specified for the five-pound can.

5.2.1.2.2 *Thirty-pound can.* Thirty pounds of the wax shall be filled into a metal pail conforming to type II, class 3 of PPP-P-704. Each pail shall have exterior and interior protective coatings. Color of the exterior coating shall acceptably match X24087 of Fed. Std. No. 595.

5.2.2 *Level B.*

5.2.2.1 *Type I.*

5.2.2.1.1 *One-gallon containers.* The wax, packaged as specified in 5.1.1.1.1, shall be packed in fiberboard shipping containers in accordance with the exterior packing requirements for domestic shipment in the Appendix of PPP-C-96.

5.2.2.1.2 *Five-gallon containers.* Five gallons of the wax shall be filled into a metal pail conforming to type I, class 2 of PPP-P-704. The closure and protective coatings shall be as specified in 5.2.1.1.2. The color of protective coating shall be that normally used in commercial practice.

5.2.2.1.3 *Fifty-five gallon containers.* Fifty-five gallons of the wax shall be filled into a metal drum conforming to type II of PPP-D-729.

5.2.2.2 *Type II.*

5.2.2.2.1 *One-pound can and four-pound can.* The wax, packaged as specified in fiberboard shipping containers in accordance with the exterior packing requirements for domestic shipment in the Appendix of PPP-C-96, except as specified in 5.2.1.2.1.

5.2.2.2.2 *Thirty-pound can.* Thirty pounds of the wax shall be filled into a metal pail conforming to type II, class 2 of PPP-P-704. Coatings and color of the pails shall be as specified in 5.2.1.2.2.

5.2.3 *Level C.* Types I and II wax, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Labeling.* Label marking shall be durably and legibly marked on the main panel of the body of each unit container by lithographing, stenciling, or printing with a black, non-corrosive, weather-resistant, waterproof ink as specified in MIL-STD-129. Each unit container shall bear clear and concise directions for use and the phrase "Do not use on asphalt, vinyl or rubber tile floors". In addition, the statement "Caution: Combustible mixture" shall appear on the label in a prominent position in not smaller than 8-point capital letters. Labeling data shall be in accordance with the Federal Hazardous Substances Labeling Act.

5.3.2 *Civil agencies.* In addition to labeling specified in 5.3.1 and any special marking required by the contract or order, unit containers and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 *Military requirements.* In addition to labeling specified in 5.3.1 and any special marking required by the contract or order, unit containers and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The waxes covered by this specification are intended for use on furniture, painted surfaces, and floors except asphalt, vinyl and rubber composition.

6.2 Ordering data. Procurement documents should specify the following:

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(*a*) Title, number and date of this specification.
(*b*) Quantity and type of wax desired.
(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 **Linoleum.** Linoleum samples may be obtained from the Chemical Specialties Manufacturer's Association, 50 East 41st Street, New York 17, New York.

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Polishing compounds, not otherwise indexed by name, including floor or furniture wax.
Carload minimum weight 36,000 pounds.

Motor:

Polishing compounds, not otherwise indexed, including floor or furniture wax.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY CUSTODIANS:

Army—GL

Navy—Sh

Air Force—MOA

OTHER INTEREST:

Army—Md

Navy—YD MC

Copies of this specification may be purchased for [illegible] cents each.

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